

Faculdade de Ciências - Departamento de Computação

Disciplina: Métodos Numéricos Computacionais

Docente: Douglas Rodrigues

Lista de Exercícios - L2

Diferenciação Numérica

Instruções:

- A lista é individual.
- A resolução deve ser feita de próprio punho (manuscrita).
- A lista deve ser entregue no formato PDF.
- O arquivo deve ser renomeado com o nome e sobrenome seguidos pela letra “L” e o número da lista.
- Deve-se utilizar letras minúsculas separados por traço baixo (underline).
- **Exemplo:** douglas_rodrigues_l2.pdf

Exercícios

1: Dada as funções, calcule o que se pede com precisão $\epsilon = 10^{-2}$:

(a) $f(x) = x^8 - 2x^7 + 5x^4 - 10x^2 + 20$, calcule o valor de $f'(1)$ e $f''(1)$.

(b) $f(x) = \sin^2(x)$ calcule o valor de $f'(\frac{\pi}{3})$ e $f''(\frac{\pi}{3})$.

(c) $f(x) = (\sin(x) * \cos(3x))^2$, calcule $f'(-2)$ e $f''(1,5)$.

(d) $f(x) = x_1^2 + 2x_2^3$, calcule $\frac{\partial f}{\partial x_1}$ no ponto $x = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$.

(e) $f(x) = \frac{x_1 + x_2^2}{x_1 x_2}$, calcule $\frac{\partial f}{\partial x_2}$ no ponto $x = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$.

(f) $f(x) = 2x_1 - 2x_1 x_2 + x_2^2$, calcule $\nabla f(x)$ no ponto $x = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$.

(g) $f(x) = 3x_1 + 2x_1 x_2 + x_2^3$, calcule $\nabla f(x)$ no ponto $x = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$.

(h) Calcule $J(x)$ no ponto $x = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$ considerando as funções:

- $f_1(x) = 2x_1 - 2x_1 x_2 + x_2^2$.
- $f_2(x) = x_1^2 + 2x_2^3$.
- $f_3(x) = 5x_1^3 + 2x_1 x_2$.

Lista de Exercícios 2 - Diferenciação Numérica

Nome: Igor dos Reis Gomes

RA: 241025265

1- $\varepsilon = 10^{-2}$

a) $f(x) = x^8 - 2x^7 + 5x^4 - 10x^2 + 20$, $f'(1) = ?$ e $f''(1) = ?$

$$i=1 \quad f'_1(1) = \frac{f(2) - f(0)}{2 \cdot 1} = \frac{60 - 20}{2} = 20$$

$$i=2 \quad f'_2(1) = \frac{f(1,5) - f(0,5)}{2 \cdot \frac{1}{2}} = \frac{14,2695 - 17,8008}{1} = -3,5313 \quad E = 6,6636$$

$$i=3 \quad f'_3(1) = \frac{f(1,25) - f(0,75)}{2 \cdot \frac{1}{4}} = \frac{13,0058 - 15,7902}{0,5} = -5,5688 \quad E = 0,3659$$

$$i=4 \quad f'_4(1) = \frac{f(1,125) - f(0,875)}{2 \cdot \frac{1}{8}} = \frac{13,3572 - 14,8329}{0,25} = -5,9028 \quad E = 0,0566$$

$$i=5 \quad f'_5(1) = \frac{f(1,0625) - f(0,9375)}{2 \cdot \frac{1}{16}} = \frac{13,6500 - 14,3970}{0,125} = -5,9760 \quad E = 0,0122$$

$$i=6 \quad f'_6(1) = \frac{f(1,0313) - f(0,9688)}{2 \cdot \frac{1}{32}} = \frac{13,8183 - 14,1929}{0,0625} = -5,9936 \quad E = 0,0029 < \varepsilon$$

$$i=1 \quad f''_1(1) = \frac{f(3) - 2f(1) + f(-1)}{4 \cdot 1^2} = \frac{2522 - 28 + 18}{4} = 628$$

$$i=2 \quad f''_2(1) = \frac{f(2) - 2f(1) + f(0)}{4 \cdot \frac{1}{2}^2} = \frac{60 - 28 + 20}{1} = 52 \quad E = 11,0769$$

$$i=3 \quad f_3''(1) = \frac{f(1,5) - 2f(1) + f(0,5)}{4 \cdot \frac{1}{4}^2} = \frac{14,2695 - 28 + 17,8008}{0,25} = 16,2812 \quad E = 2,1939$$

$$h = \frac{1}{4}$$

$$i=4 \quad f_4''(1) = \frac{f(1,25) - 2f(1) + f(0,75)}{4 \cdot \frac{1}{8}^2} = \frac{13,0058 - 28 + 15,7902}{0,0625} = 12,736 \quad E = 0,2784$$

$$h = \frac{1}{8}$$

$$i=5 \quad f_5''(1) = \frac{f(1,125) - 2f(1) + f(0,875)}{4 \cdot \frac{1}{16}^2} = \frac{13,3572 - 28 + 14,8329}{0,015625} = 12,1664 \quad E = 0,0468$$

$$h = \frac{1}{16}$$

$$i=6 \quad f_6''(1) = \frac{f(1,0625) - 2f(1) + f(0,9375)}{4 \cdot \frac{1}{32}^2} = \frac{13,6500 - 28 + 14,3970}{0,0039} = 12,0513 \quad E = 0,0096 < \varepsilon$$

$$h = \frac{1}{32}$$

$$f'(1) = -5,9936 \quad \text{e} \quad f''(1) = 12,0513 //$$

b) $f(x) = \sin^2 x$, $f'(\pi/3) = ?$ e $f''(\pi/3) = ?$

$$i=1 \quad f_1'(\pi/3) = \frac{f(\pi/3+1) - f(\pi/3-1)}{2 \cdot 1} = \frac{0,7897 - 0,0022}{2} = 0,3938$$

$$h = 1$$

$$i=2 \quad f_2'(\pi/3) = \frac{f(\pi/3+1/2) - f(\pi/3-1/2)}{2 \cdot 1/2} = \frac{0,9994 - 0,2707}{1} = 0,7287 \quad E = 0,3349$$

$$h = 1/2$$

$$i=3 \quad f_3'(\pi/3) = \frac{f(\pi/3+1/4) - f(\pi/3-1/4)}{2 \cdot 1/4} = \frac{0,9270 - 0,5118}{0,5} = 0,8304 \quad E = 0,1017$$

$$h = 1/4$$

$$i=4 \quad f_4'(\pi/3) = \frac{f(\pi/3+1/8) - f(\pi/3-1/8)}{2 \cdot 1/8} = \frac{0,8494 - 0,6351}{0,25} = 0,8572 \quad E = 0,0268$$

$$h = 1/8$$

$$i=5 \quad f_5'(\pi/3) = \frac{f(\pi/3+1/16) - f(\pi/3-1/16)}{2 \cdot 1/16} = \frac{0,8020 - 0,6941}{0,125} = 0,8632 \quad E = 0,006 < \varepsilon$$

$$h = 1/16$$

$$i=1 \quad f_1''(x_3) = \frac{f(x_3+2) - 2f(x_3) + f(x_3-2)}{4 \cdot 1^2} = \frac{0,0089 - 1,5 + 0,6643}{4} = -0,2067$$

$$h=1$$

$$i=2 \quad f_2''(x_3) = \frac{f(x_3+1) - 2f(x_3) + f(x_3-1)}{4 \cdot \frac{1}{2}^2} = \frac{0,7897 - 1,5 + 0,0022}{1} = -0,7081 \quad E = 0,5014$$

$$h=\frac{1}{2}$$

$$i=3 \quad f_3''(x_3) = \frac{f(x_3+0,5) - 2f(x_3) + f(x_3-0,5)}{4 \cdot \frac{1}{4}^2} = \frac{0,9994 - 1,5 + 0,2707}{0,25} = -0,9196 \quad E = 0,2115$$

$$h=\frac{1}{4}$$

$$i=4 \quad f_4''(x_3) = \frac{f(x_3+0,25) - 2f(x_3) + f(x_3-0,25)}{4 \cdot \frac{1}{8}^2} = \frac{0,9270 - 1,5 + 0,5118}{0,0625} = -0,9792 \quad E = 0,0596$$

$$h=\frac{1}{8}$$

$$i=5 \quad f_5''(x_3) = \frac{f(x_3+0,125) - 2f(x_3) + f(x_3-0,125)}{4 \cdot \frac{1}{16}^2} = \frac{0,8494 - 1,5 + 0,6351}{0,015625} = -0,992 \quad E = 0,0128$$

$$h=\frac{1}{16}$$

$$i=6 \quad f_6''(x_3) = \frac{f(x_3+\frac{1}{32}) - 2f(x_3) + f(x_3-\frac{1}{32})}{4 \cdot \frac{1}{32}^2} = \frac{0,8020 - 1,5 + 0,6941}{\frac{1}{256}} = -0,9984 \quad E = 0,0064$$

$$h=\frac{1}{32}$$

$$f'(x_3) = 0,8632 \quad \text{e} \quad f''(x_3) = -0,9984 //$$

c) $(\sin x \cdot \cos 3x)^2$, $f'(-2) = ?$ e $f''(1,5) = ?$

$$i=1 \quad f_1'(-2) = \frac{f(-1) - f(-3)}{2 \cdot 1} = \frac{0,6940 - 0,0165}{2} = 0,3388$$

$$h=1$$

$$i=2 \quad f_2'(-2) = \frac{f(-1,5) - f(-2,5)}{2 \cdot \frac{1}{2}} = \frac{0,0442 - 0,0430}{1} = 0,0012 \quad E = 0,3376$$

$$h=\frac{1}{2}$$

$$i=3 \quad f_3'(-2) = \frac{f(-1,75) - f(-2,25)}{2 \cdot \frac{1}{4}} = \frac{0,2539 - 0,4828}{0,5} = -0,4578 \quad E = 0,459$$

$$h=\frac{1}{4}$$

$$i=4 \quad h=\frac{1}{8} \quad f'_4(-2) = \frac{f(-1,875) - f(-2,125)}{2 \cdot \frac{1}{8}} = \frac{0,5637 - 0,7170}{0,25} = -0,5892 \quad E = 0,1314$$

$$i=5 \quad h=\frac{1}{16} \quad f'_5(-2) = \frac{f(-1,9375) - f(-2,0625)}{2 \cdot \frac{1}{16}} = \frac{0,6922 - 0,7700}{0,125} = -0,6224 \quad E = 0,0332$$

$$i=6 \quad h=\frac{1}{32} \quad f'_6(-2) = \frac{f(-1,9688) - f(-2,0313)}{2 \cdot \frac{1}{32}} = \frac{0,7347 - 0,7741}{0,0625} = -0,64 \quad E = 0,0176$$

$$i=7 \quad h=\frac{1}{64} \quad f'_7(-2) = \frac{f(-1,9844) - f(-2,0156)}{2 \cdot \frac{1}{64}} = \frac{0,7504 - 0,7702}{\frac{1}{32}} = -0,6336 \quad E = 0,0064$$

$$i=1 \quad h=1 \quad f''_{1(1,5)} = \frac{f(3,5) - 2f(1,5) + f(-0,5)}{4 \cdot 1^2} = \frac{0,0278 - 0,0884 + 0,0012}{4} = -0,0149$$

$$i=2 \quad h=\frac{1}{2} \quad f''_{2(1,5)} = \frac{f(2,5) - 2f(1,5) + f(0,5)}{4 \cdot \frac{1}{2}^2} = \frac{0,0430 - 0,0884 + 0,0012}{1} = -0,0442 \quad E = 0,0293$$

$$i=3 \quad h=\frac{1}{4} \quad f''_{3(1,5)} = \frac{f(2) - 2f(1,5) + f(1)}{4 \cdot \frac{1}{4}^2} = \frac{0,7623 - 0,0884 + 0,6940}{0,25} = 5,9716 \quad E = 1,0081$$

$$i=4 \quad h=\frac{1}{8} \quad f''_{4(1,5)} = \frac{f(1,75) - 2f(1,5) + f(1,25)}{4 \cdot \frac{1}{8}^2} = \frac{0,2539 - 0,0884 + 0,6064}{0,0625} = 12,3504 \quad E = 0,5570$$

$$i=5 \quad h=\frac{1}{16} \quad f''_{5(1,5)} = \frac{f(1,625) - 2f(1,5) + f(1,375)}{4 \cdot \frac{1}{16}^2} = \frac{0,0261 - 0,0884 + 0,2955}{4 \cdot \frac{1}{16}^2} = 14,9248 \quad E = 0,1725$$

$$i=6 \quad h=\frac{1}{32} \quad f''_{6(1,5)} = \frac{f(1,5625) - 2f(1,5) + f(1,4375)}{4 \cdot \frac{1}{32}^2} = \frac{0,0006 - 0,0884 + 0,1489}{4 \cdot \frac{1}{32}^2} = 15,6416 \quad E = 0,0458$$

0,01

$$i=7 \quad f_7''(1,5) = \frac{f(1,5313) - 2f(1,5) + f(1,4687)}{4 \cdot \frac{1}{64}^2} = \frac{0,0140 - 0,0884 + 0,0900}{4 \cdot \frac{1}{64}^2} = 15,9744 \quad E = 0,0208$$

$$h = \frac{1}{64}$$

$$i=8 \quad f_8''(1,5) = \frac{f(1,5156) - 2f(1,5) + f(1,4844)}{4 \cdot \frac{1}{128}^2} = \frac{0,0271 - 0,0884 + 0,0652}{4 \cdot \frac{1}{128}^2} = 15,9744 \quad E = 0 < \varepsilon$$

$$h = \frac{1}{128}$$

$$f'(1,2) = -0,6336 \text{ e } f''(1,5) = 15,9744 //$$

$$d) f(x) = x_1^2 + 2x_2^3, \quad \frac{\partial f}{\partial x_1}, \quad x = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$$

$$i=1 \quad \frac{\partial f}{\partial x_1} = \frac{f(3,-2) - f(1,-2)}{2 \cdot 1} = \frac{-7 + 15}{2} = 4$$

$$h=1$$

$$i=2 \quad \frac{\partial f}{\partial x_1} = \frac{f(2,5,-2) - f(1,5,-2)}{2 \cdot \frac{1}{2}} = \frac{-9,75 + 13,75}{1} = 4 //$$

$$h = \frac{1}{2} \quad E = 0 < \varepsilon$$

$$e) f(x) = \frac{x_1 + x_2^2}{x_1 x_2}, \quad \frac{\partial f}{\partial x_2}, \quad x = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$$

$$i=1 \quad \frac{\partial f}{\partial x_2} = \frac{f(-2,3) - f(-2,1)}{2 \cdot 1} = \frac{-1,6667 - 0,5}{2} = -1,0834$$

$$h=1$$

$$i=2 \quad \frac{\partial f}{\partial x_2} = \frac{f(-2,2,5) - f(-2,1,5)}{2 \cdot \frac{1}{2}} = \frac{-0,85 + 0,0833}{1} = -0,7667 \quad E = 0,3167$$

$$h = \frac{1}{2}$$

$$i=3 \quad \frac{\partial f}{\partial x_2} = \frac{f(-2,2,25) - f(-2,1,75)}{2 \cdot \frac{1}{4}} = \frac{-0,6806 + 0,3036}{\frac{1}{2}} = -0,754 \quad E = 0,0127$$

$$h = \frac{1}{4}$$

$$i=4 \quad \frac{\partial f}{\partial x_2} = \frac{f(1,2,2,125) - f(1,2,1,875)}{2 \cdot \frac{1}{8}} = \frac{-0,5819 + 0,4042}{\frac{1}{4}} = -0,7508 // \quad E = 0,0032 < \varepsilon$$

$$f) f(x) = 2x_1 - 2x_1x_2 + x_2^2, \quad \nabla f(x), \quad x = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$$

$$i=1 \quad \frac{\partial f}{\partial x_1} = \frac{f(8,3) - f(6,3)}{2 \cdot 1} = \frac{-23 + 15}{2} = -4$$

$$i=2 \quad \frac{\partial f}{\partial x_1} = \frac{f(7,5,3) - f(6,5,3)}{2 \cdot \frac{1}{2}} = \frac{-21 + 17}{1} = -4 // \quad E = 0 < \varepsilon$$

$$i=1 \quad \frac{\partial f}{\partial x_2} = \frac{f(7,4) - f(7,2)}{2 \cdot 1} = \frac{-26 + 10}{2} = -8$$

$$i=2 \quad \frac{\partial f}{\partial x_2} = \frac{f(7,3,5) - f(7,2,5)}{2 \cdot \frac{1}{2}} = \frac{-22,75 + 14,75}{1} = -8 // \quad E = 0 < \varepsilon$$

$$\nabla f(x_1, x_2) = \begin{pmatrix} -4 \\ -8 \end{pmatrix} //$$

$$g) f(x) = 3x_1 + 2x_1x_2 + x_2^3, \quad \nabla f(x), \quad x = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

$$i=1 \quad \frac{\partial f}{\partial x_1} = \frac{f(2,4) - f(0,4)}{2 \cdot 1} = \frac{86 - 64}{2} = 11$$

$$i=2 \quad \frac{\partial f}{\partial x_1} = \frac{f(1,5,4) - f(0,5,4)}{2 \cdot \frac{1}{2}} = \frac{80,5 - 69,5}{1} = 11 // \quad E = 0 < \varepsilon$$

$$i=1 \quad \frac{\partial f}{\partial x_2} = \frac{f(1,5) - f(1,3)}{2 \cdot 1} = \frac{138 - 36}{2} = 51$$

$$i=2 \quad \frac{\partial f}{\partial x_2} = \frac{f(1,4,5) - f(1,3,5)}{2 \cdot \frac{1}{2}} = \frac{103,125 - 52,875}{1} = 50,25 \quad E = 0,0149$$

$$i=3 \quad \frac{\partial f}{\partial x_2} = \frac{f(1,4,25) - f(1,3,25)}{2 \cdot \frac{1}{4}} = \frac{88,2656 - 63,2344}{\frac{1}{2}} = 50,0624 \quad E = 0,0037 < \epsilon$$

$$\nabla f(x_1, x_2) = \begin{pmatrix} 11 \\ 50,0624 \end{pmatrix}$$

h)

$$J(x) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} \end{bmatrix}, \quad \epsilon = 10^{-2}, \quad x = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

$$f_1(x) = 2x_1 - 2x_1x_2 + x_2^2$$

$$f_2(x) = x_1^2 + 2x_2^3$$

$$f_3(x) = 5x_1^3 + 2x_1x_2$$

$$i=1 \quad \frac{\partial f_1}{\partial x_1} = \frac{f(2,5) - f(0,5)}{2 \cdot 1} = \frac{9 - 25}{2} = -8$$

$$i=2 \quad \frac{\partial f_1}{\partial x_1} = \frac{f(1,5,5) - f(0,5,5)}{2 \cdot \frac{1}{2}} = \frac{13 - 21}{1} = -8 \quad E = 0 < \epsilon$$

$$i=1 \quad \frac{\partial f_1}{\partial x_2} = \frac{f(1,6) - f(1,4)}{2 \cdot 1} = \frac{26 - 10}{2} = 8$$

$$i=2 \quad \frac{\partial f_1}{\partial x_2} = \frac{f(1,5,5) - f(1,4,5)}{2 \cdot \frac{1}{2}} = \frac{21,25 - 13,25}{1} = 8 \quad E = 0 < \epsilon$$

$$\begin{array}{l} i=1 \\ h=1 \end{array} \quad \frac{\partial f_2}{\partial x_1} = \frac{f(1,2,5) - f(1,0,5)}{2 \cdot 1} = \frac{254 - 250}{2} = 2$$

$$\begin{array}{l} i=2 \\ h=1/2 \end{array} \quad \frac{\partial f_2}{\partial x_1} = \frac{f(1,1,5,5) - f(1,0,5,5)}{2 \cdot 1/2} = \frac{252,25 - 250,25}{1} = 2 // \quad E = 0 < \epsilon$$

$$\begin{array}{l} i=1 \\ h=1 \end{array} \quad \frac{\partial f_2}{\partial x_2} = \frac{f(1,1,6) - f(1,1,4)}{2 \cdot 1} = \frac{433 - 129}{2} = 152$$

$$\begin{array}{l} i=2 \\ h=1/2 \end{array} \quad \frac{\partial f_2}{\partial x_2} = \frac{f(1,1,5,5) - f(1,1,4,5)}{2 \cdot 1/2} = \frac{333,75 - 183,25}{1} = 150,5 \quad E = 0,0100$$

$$\begin{array}{l} i=3 \\ h=1/4 \end{array} \quad \frac{\partial f_2}{\partial x_2} = \frac{f(1,1,5,25) - f(1,1,4,75)}{2 \cdot 1/4} = \frac{290,4063 - 215,3438}{1/2} = 150,125 // \quad E = 0,0025 < \epsilon$$

$$\begin{array}{l} i=1 \\ h=1 \end{array} \quad \frac{\partial f_3}{\partial x_1} = \frac{f(1,2,5) - f(1,0,5)}{2 \cdot 1} = \frac{60 - 0}{2} = 30$$

$$\begin{array}{l} i=2 \\ h=1/2 \end{array} \quad \frac{\partial f_3}{\partial x_1} = \frac{f(1,1,5,5) - f(1,0,5,5)}{2 \cdot 1/2} = \frac{31,875 - 5,625}{1} = 26,25 \quad E = 0,1429$$

$$\begin{array}{l} i=3 \\ h=1/4 \end{array} \quad \frac{\partial f_3}{\partial x_1} = \frac{f(1,1,25,5) - f(1,0,75,5)}{2 \cdot 1/4} = \frac{22,2656 - 9,6094}{1/2} = 25,3124 \quad E = 0,0370$$

$$\begin{array}{l} i=4 \\ h=1/8 \end{array} \quad \frac{\partial f_3}{\partial x_1} = \frac{f(1,1,125,5) - f(1,0,875,5)}{2 \cdot 1/8} = \frac{18,3691 - 12,0996}{1/4} = 25,078 // \quad E = 0,0093 < \epsilon$$

$$\begin{array}{l} i=1 \\ h=1 \end{array} \quad \frac{\partial f_3}{\partial x_2} = \frac{f(1,1,6) - f(1,1,4)}{2 \cdot 1} = \frac{17 - 13}{2} = 2$$

$$i=2 \quad \frac{\partial f_3}{\partial x_2} = \frac{f(1,5,5) - f(1,4,5)}{2 \cdot \frac{1}{2}} = \frac{16 - 14}{1} = 2 //$$

$$h = \frac{1}{2} \quad E = 0 < \epsilon$$

$$J(x) = \begin{bmatrix} -8 & 8 \\ 2 & 150,125 \\ 25,078 & 2 \end{bmatrix} //$$

